



DELIVERING THE FUTURE OF HIGH-PERFORMANCE COMPUTING

DR. LISA SU

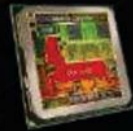
August 19, 2019

OUR HERITAGE



First to break the
historic 1GHz
barrier

2000



First to break teraflops performance barrier

2006



World's
first APU

2011



First 32-Core x86 single socket server CPU

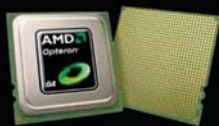
2017



World's first 7nm
datacenter GPU

2018

2019



World's first x86-64 bit architecture



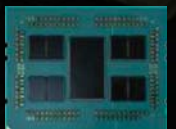
First to break
1GHz GPU barrier



Inside every major
gaming console



World's first 32 core enthusiast desktop processor



First 7nm
chiplet design

THE WORKLOADS OF THE FUTURE REQUIRE INCREDIBLE AMOUNTS OF COMPUTE POWER

HIGH PERFORMANCE COMPUTING



CLOUD, HYPERSCALE & VIRTUALIZATION



MACHINE INTELLIGENCE



BIG DATA ANALYTICS



IMMERSIVE & INSTINCTIVE COMPUTING



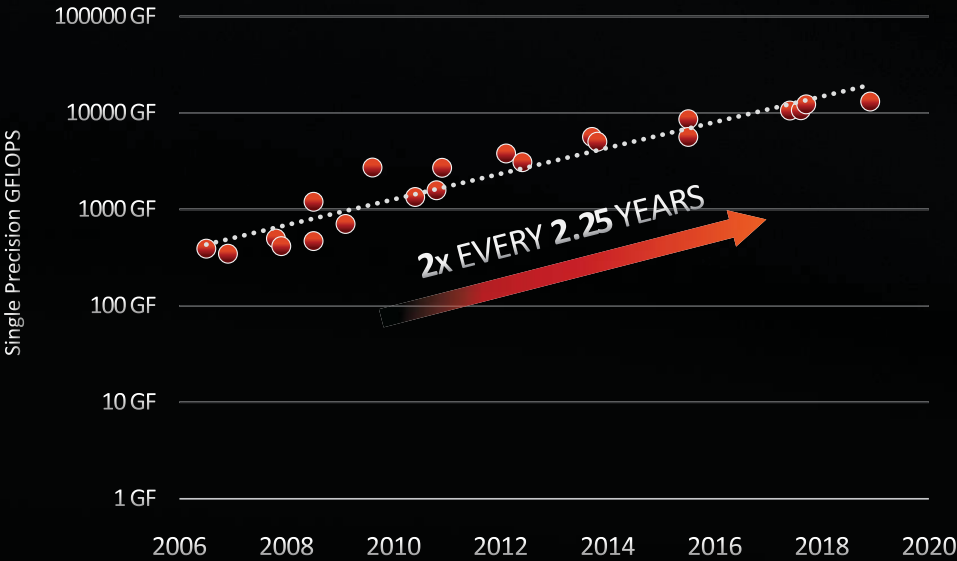
SOFTWARE-DEFINED STORAGE



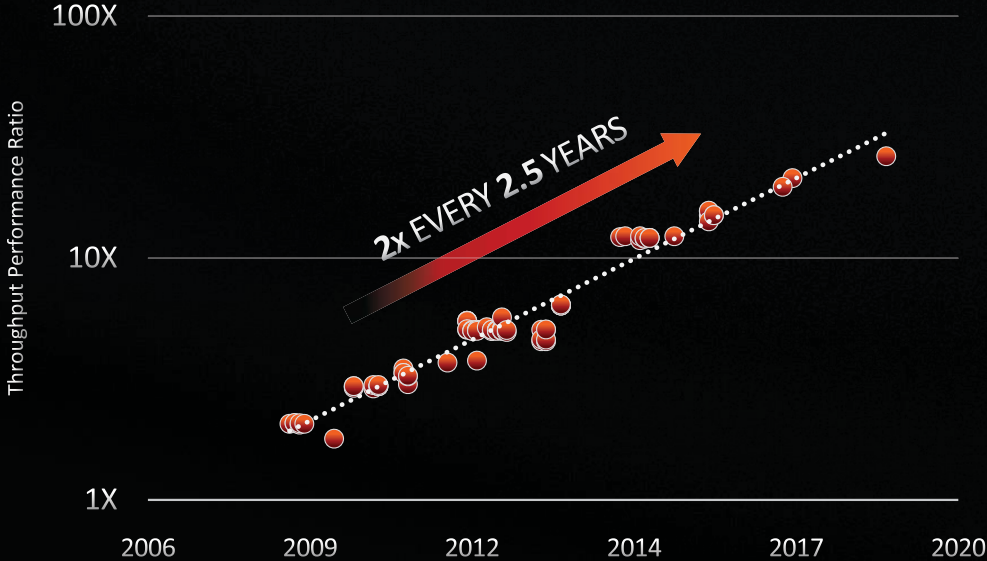
Authorized licensee of AMD. All rights reserved. Performance based on SPECint_rate2006 2P Server. Performance may vary. AMD, the AMD logo, and "AMD" are trademarks of AMD. All other trademarks are the property of their respective owners. Conditions apply.

GPU AND CPU PERFORMANCE TRENDS

GPU Single Precision Floating Point Operations Per Second Trend



Specint®_rate2006 2P Server Performance Trend Over Time

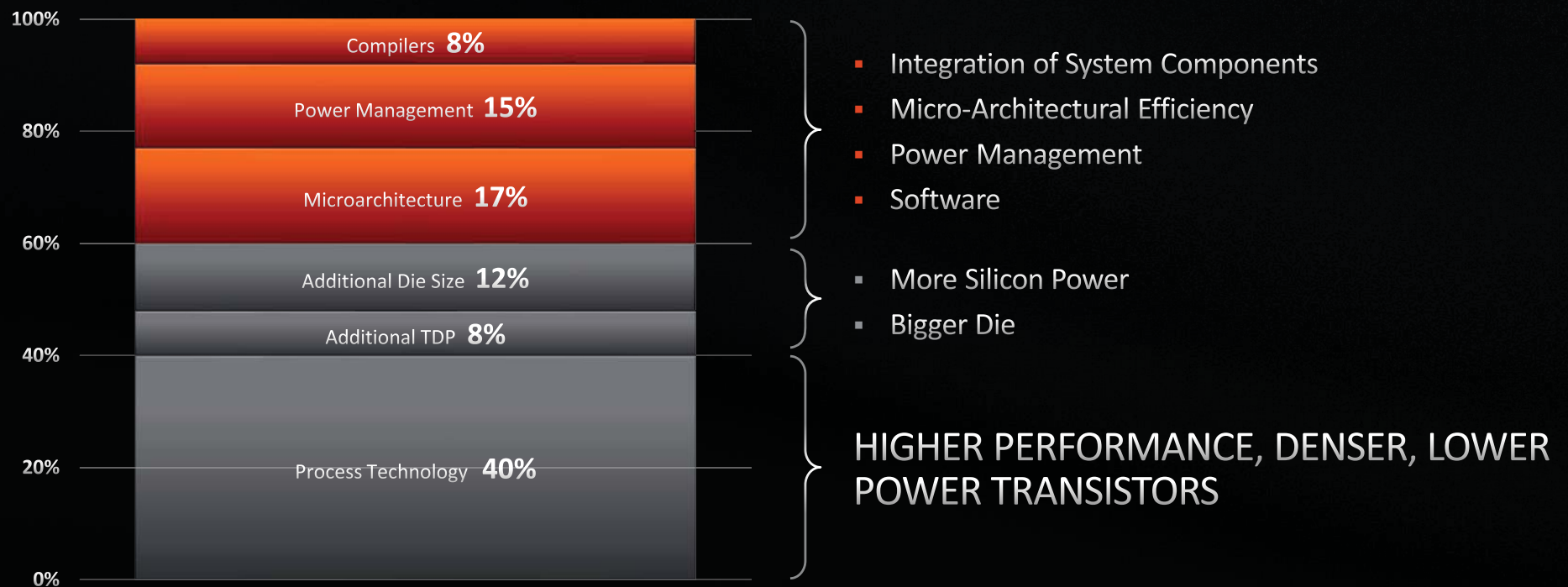


CONSISTENT AND EXPONENTIAL GPU/CPU PERFORMANCE GAINS

Technology Energy Efficiency and Density Across Process Nodes



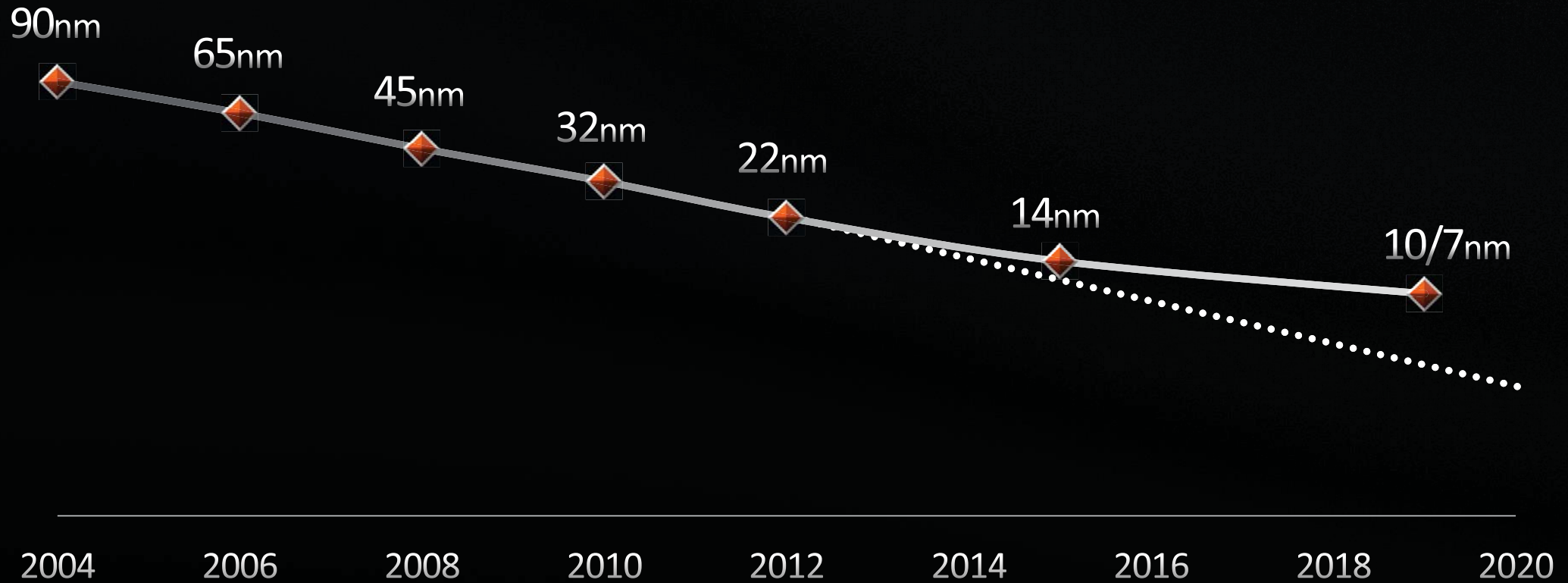
PERFORMANCE GAINS OVER THE PAST DECADE



ELEMENTS OF 2x IN 2.5 YEAR PERFORMANCE GAIN OVER THE PAST DECADE

BARRIERS TO CONTINUED PERFORMANCE IMPROVEMENT

MOORE'S LAW KEEPS SLOWING

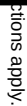


AMD Internal



6 conditions apply.

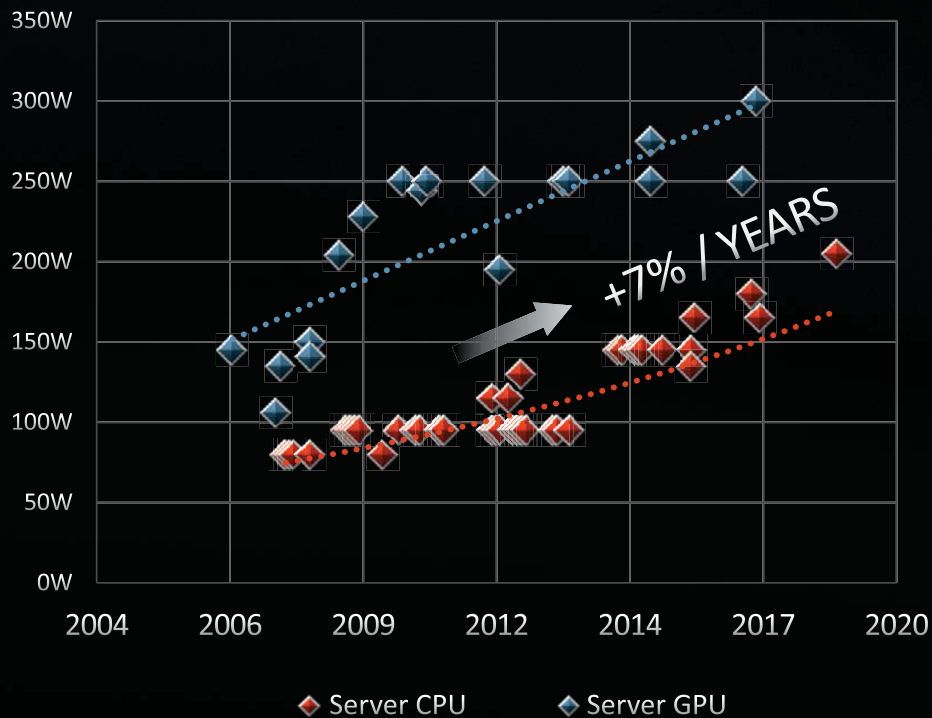
ctions apply.



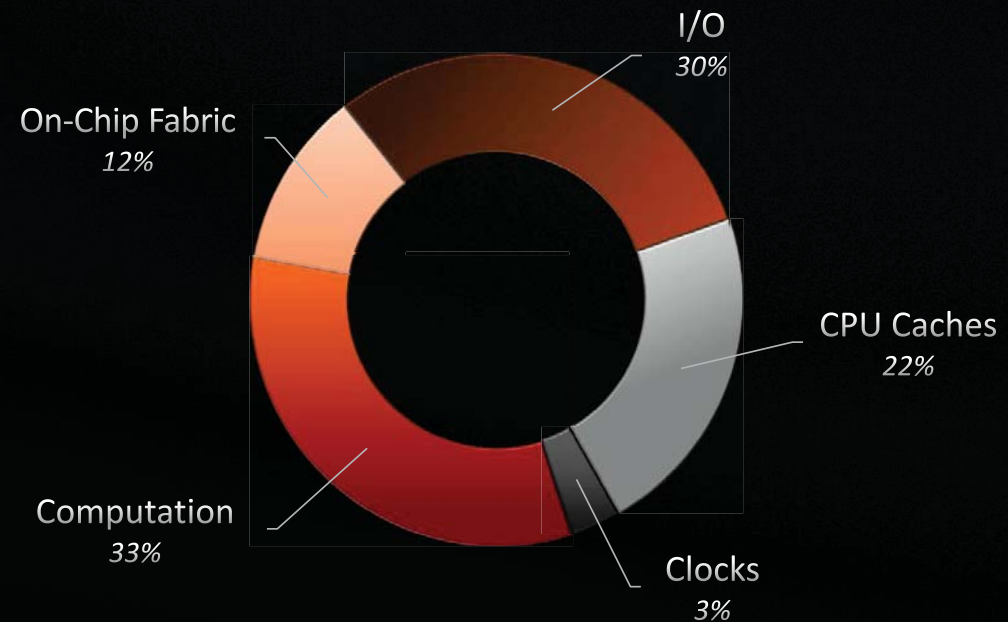
ctions apply.

SOC POWER TREND

Thermal Design Power Over Time
in Server CPUs and GPUs

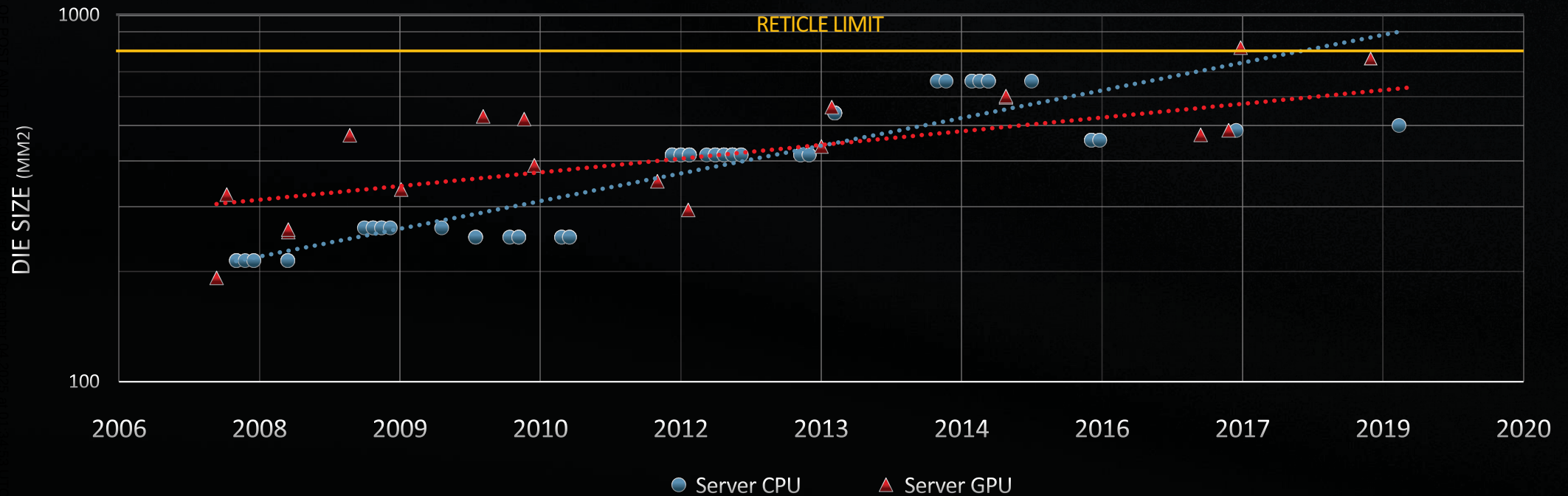


Estimated Typical Server
Power Breakdown 2018



DIE SIZE TREND

Die Size Increases Over Time in Server CPUs and GPUs

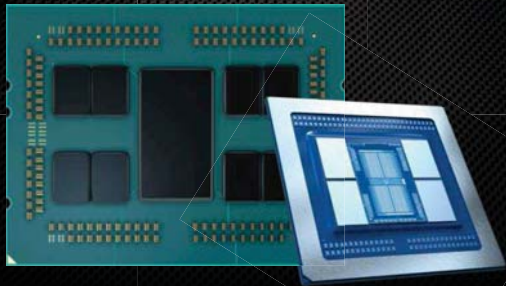


DIE SIZES INCREASING AT AN UNSUSTAINABLE RATE

NEW APPROACHES TO EXTEND PERFORMANCE GAINS

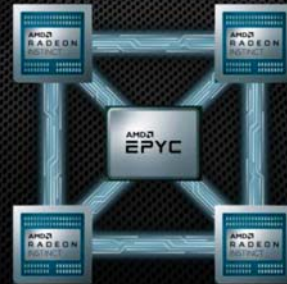
DELIVERING HIGH PERFORMANCE

SILICON



Efficient, Power Optimized Designs
with Innovative Architecture

SYSTEM



Interconnects, Memory and
Topologies Balanced to Address the
Key Performance Bottlenecks

SOFTWARE

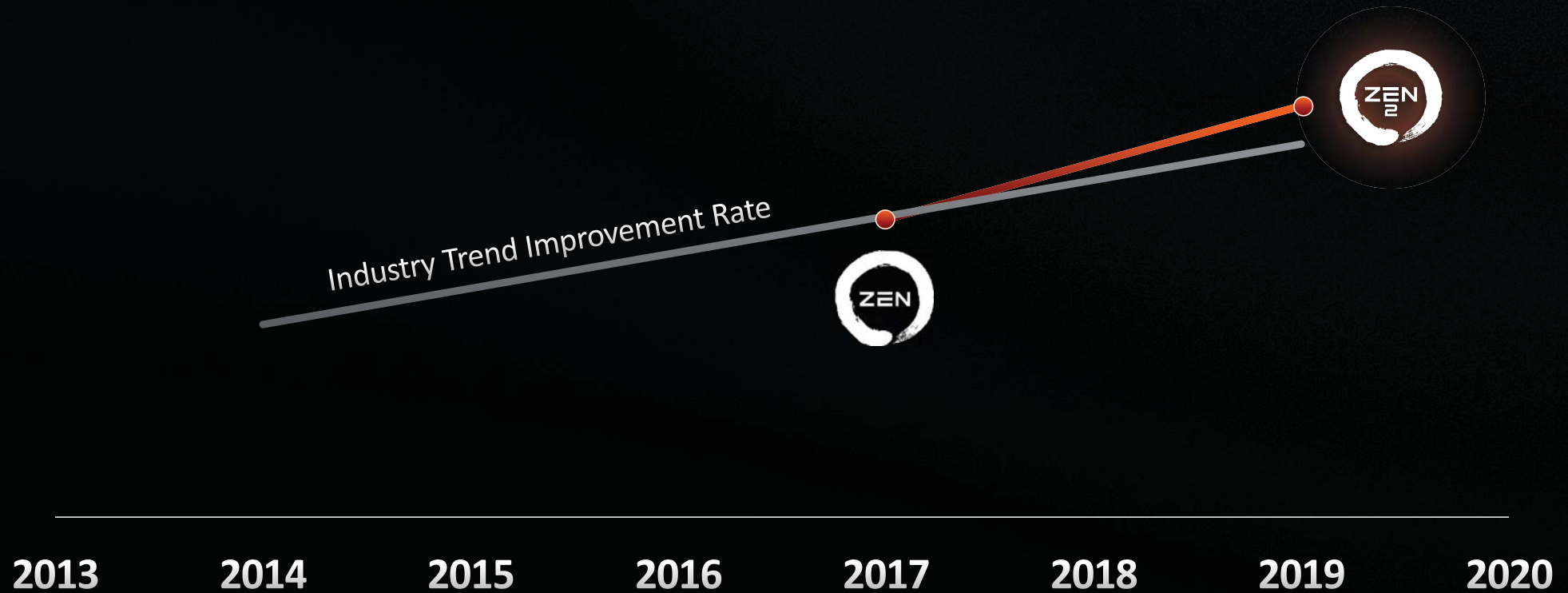
OpenMP



Co-Optimized Software to
Accelerate System Capabilities

SILICON ADVANCES

CORE DESIGN AND IPC



15% HIGHER INSTRUCTIONS PER CLOCK

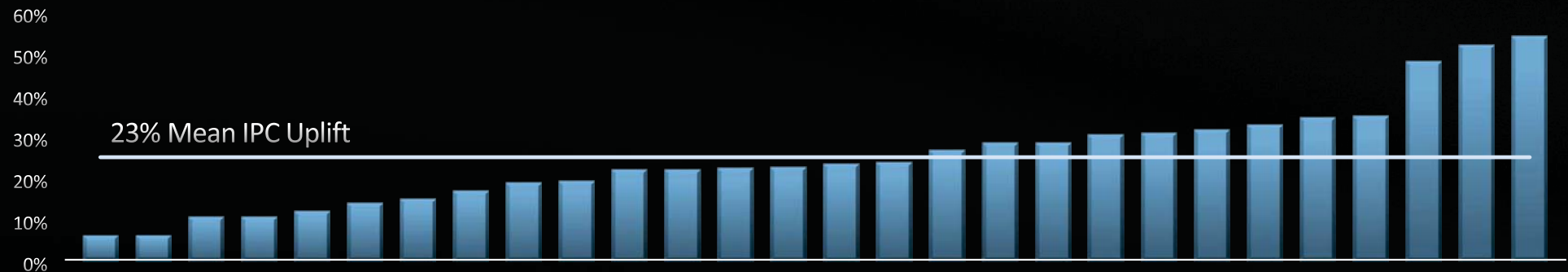
HIGHER IPC FOR SERVER WORKLOADS

2ND GEN EPYC™ SINGLE THREAD IPC UPLIFT



Components of SPEC CPU®2017_int_2017 and 2006 at ISO Frequency

2ND GEN EPYC™ 32-CORE 64-THREAD IPC UPLIFT



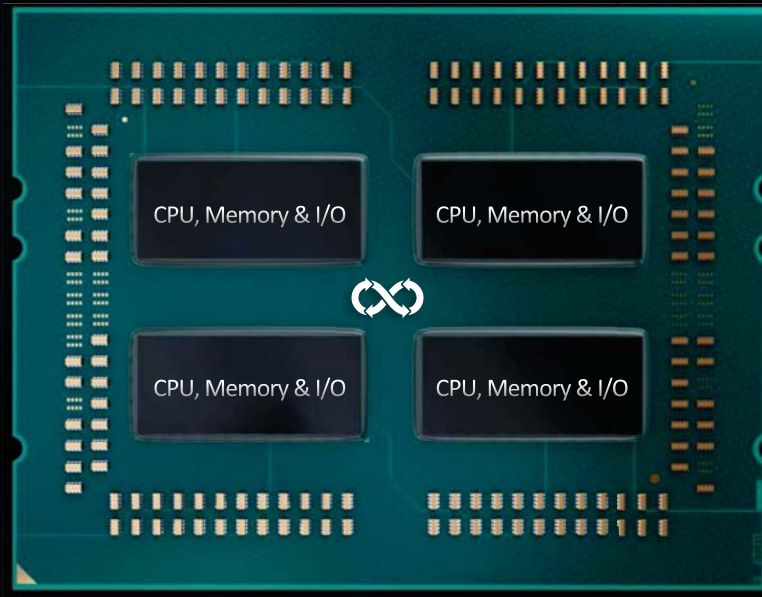
Components of of SPEC CPU®2017_int_2017 and Selection of Server Workloads

See endnotes EPYC-07, ROM-236



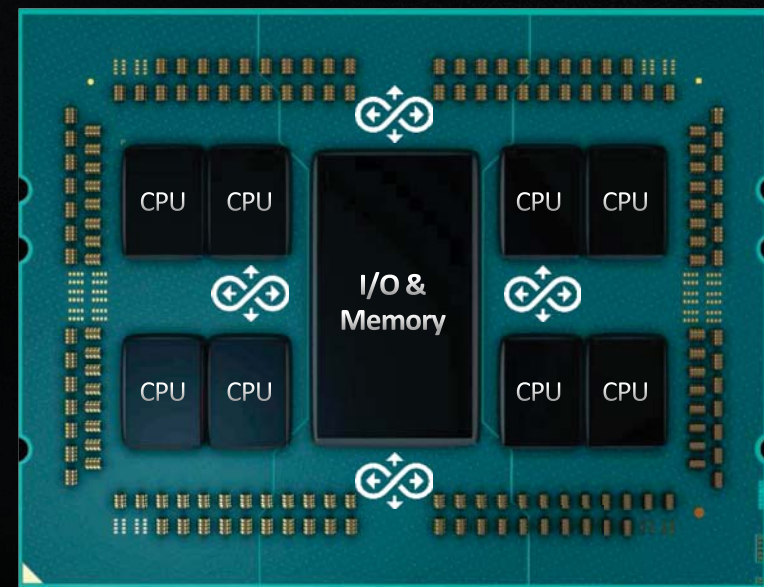
MULTI-CHIP ARCHITECTURES

1ST GENERATION



Four SOC's Interconnected
via 1st Gen AMD Infinity Architecture

2ND GENERATION



Eight 7nm Chiplet CPUs and One 12nm Chiplet I/O
Interconnected via 2nd Gen AMD Infinity Architecture

Each IP in its Optimal
Process Technology

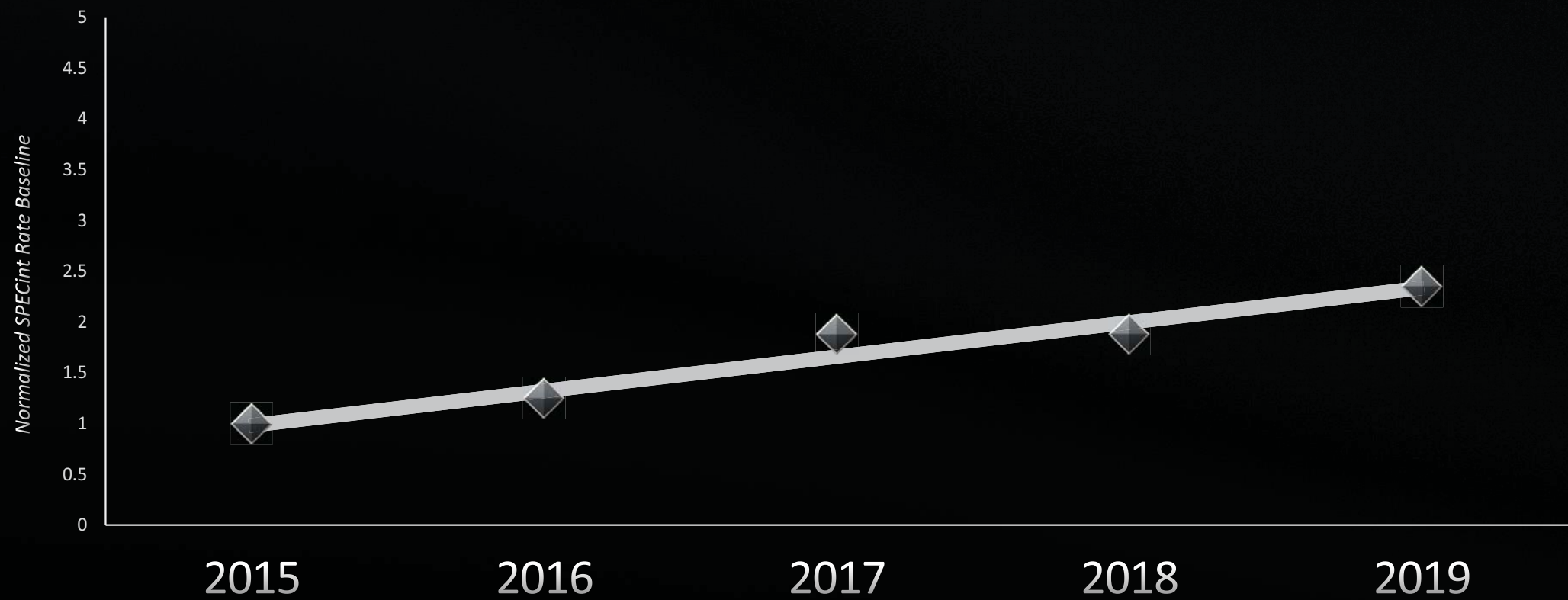
Distributed
Control

I/O Die and CPU Die Optimizes
Latency and Power

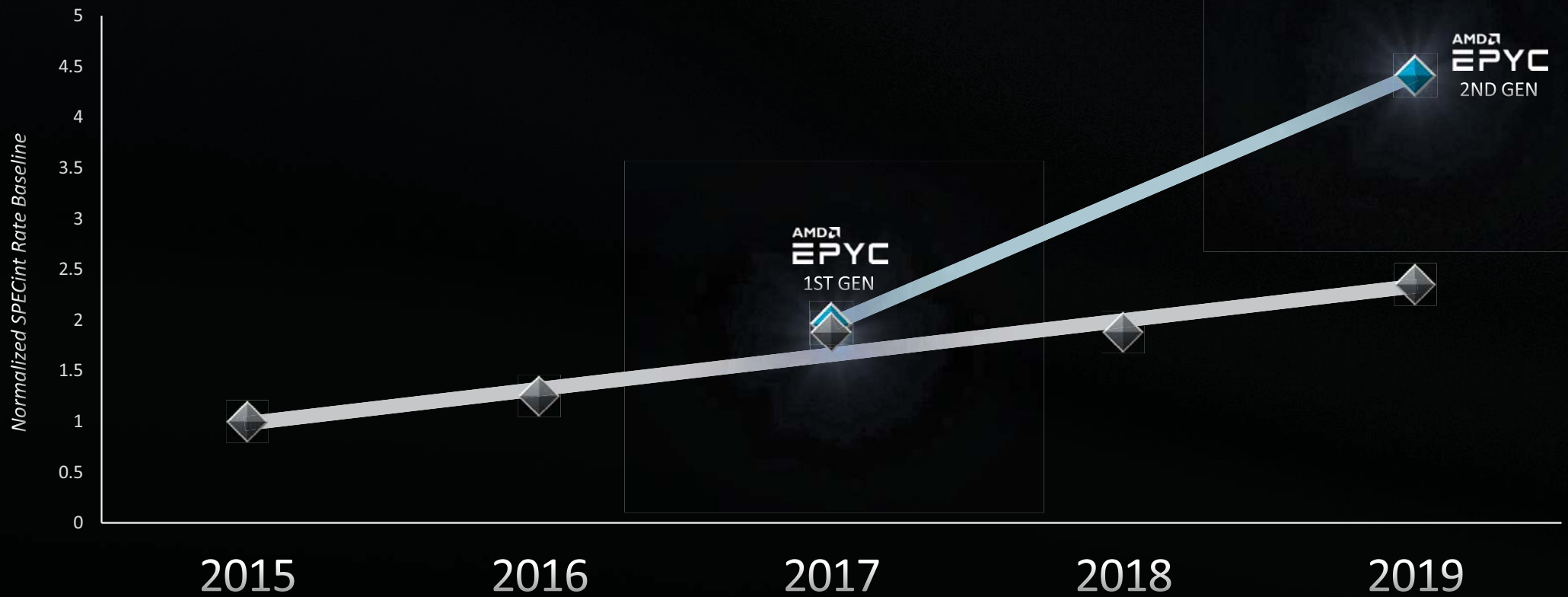
Flexible and More Unified
Memory Architecture

HISTORICAL DATACENTER PERFORMANCE

MAINSTREAM 2S SERVER PERFORMANCE



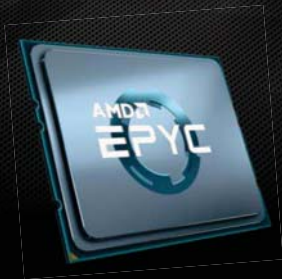
TAKING AN EPYC™ LEAP IN DATACENTER PERFORMANCE



SYSTEM ADVANCES

ACCELERATED COMPUTING PLATFORMS

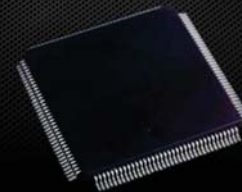
CPUs



GPUs



FPGAs/Accelerators



High Speed Interconnects



OPTIMIZING SYSTEM PERFORMANCE WITH ACCELERATED COMPUTING

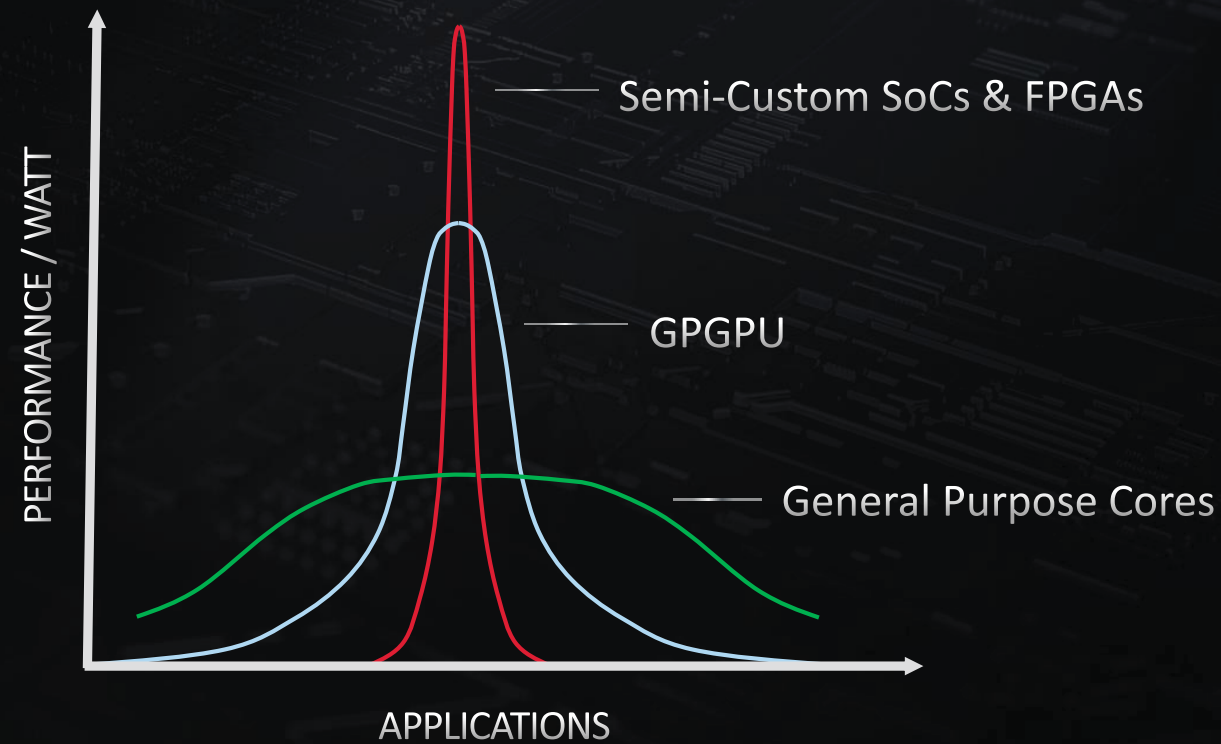
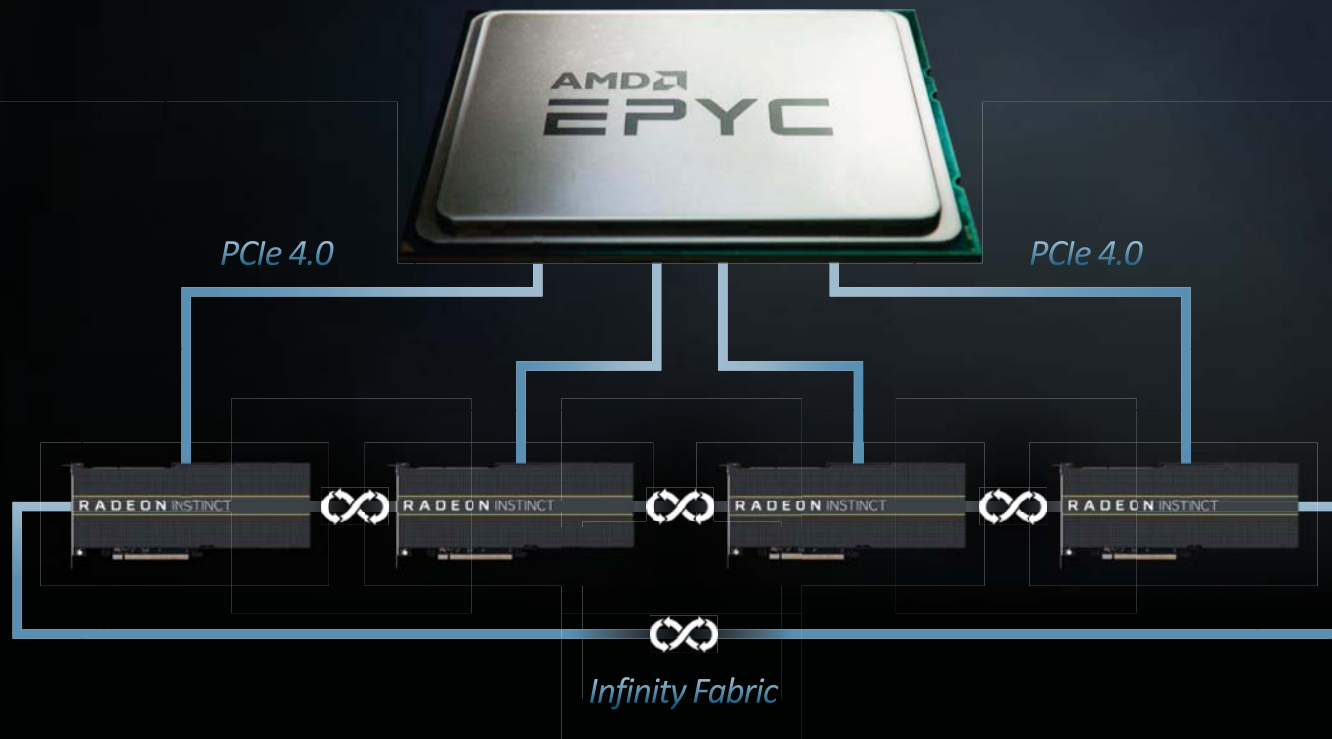


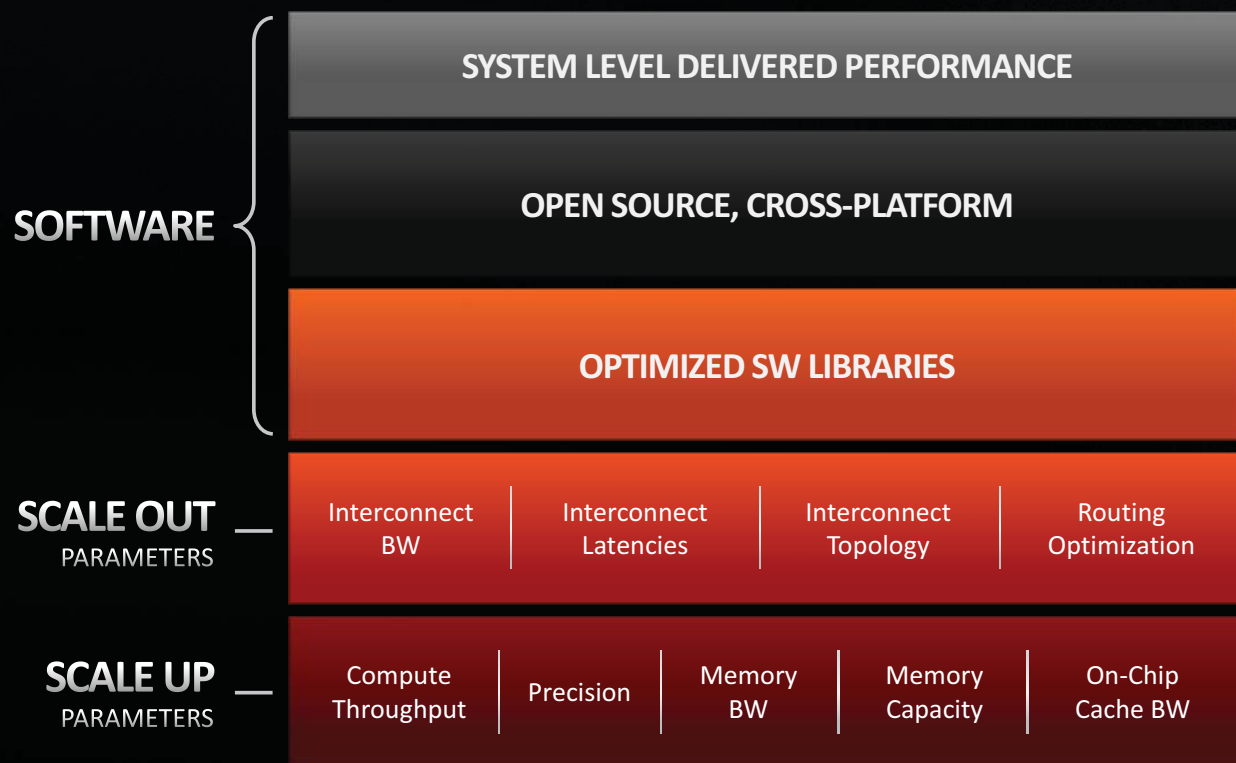
CHART FOR ILLUSTRATIVE PURPOSES

HIGH-PERFORMANCE HETEROGENEOUS PLATFORMS



SOFTWARE ADVANCES

SOFTWARE TIES IT TOGETHER



- Standard, open-source, cross-platform programming languages
- Library optimization
- Profilers and debuggers for development on AMD hardware

THE WORKLOADS OF THE FUTURE REQUIRE INCREDIBLE AMOUNTS OF COMPUTE POWER

HIGH PERFORMANCE COMPUTING



CLOUD, HYPERSCALE & VIRTUALIZATION



MACHINE INTELLIGENCE



BIG DATA ANALYTICS



IMMERSIVE & INSTINCTIVE COMPUTING



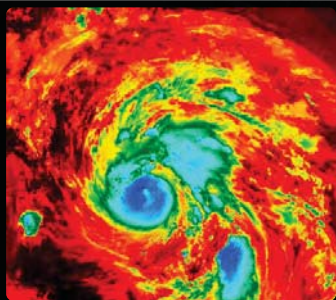
SOFTWARE-DEFINED STORAGE



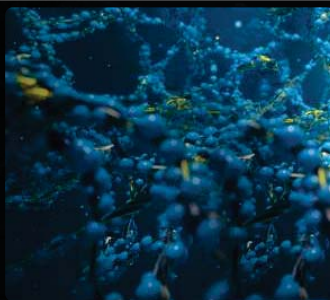
HIGH-PERFORMANCE COMPUTING APPLICATIONS



SPACE EXPLORATION



CLIMATE
CHANGE



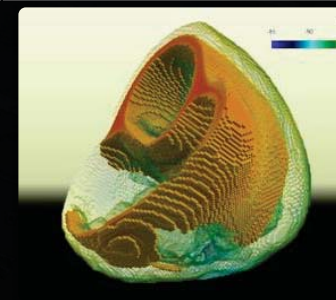
CHEMICAL SCIENCES



ENERGY
SOLUTIONS



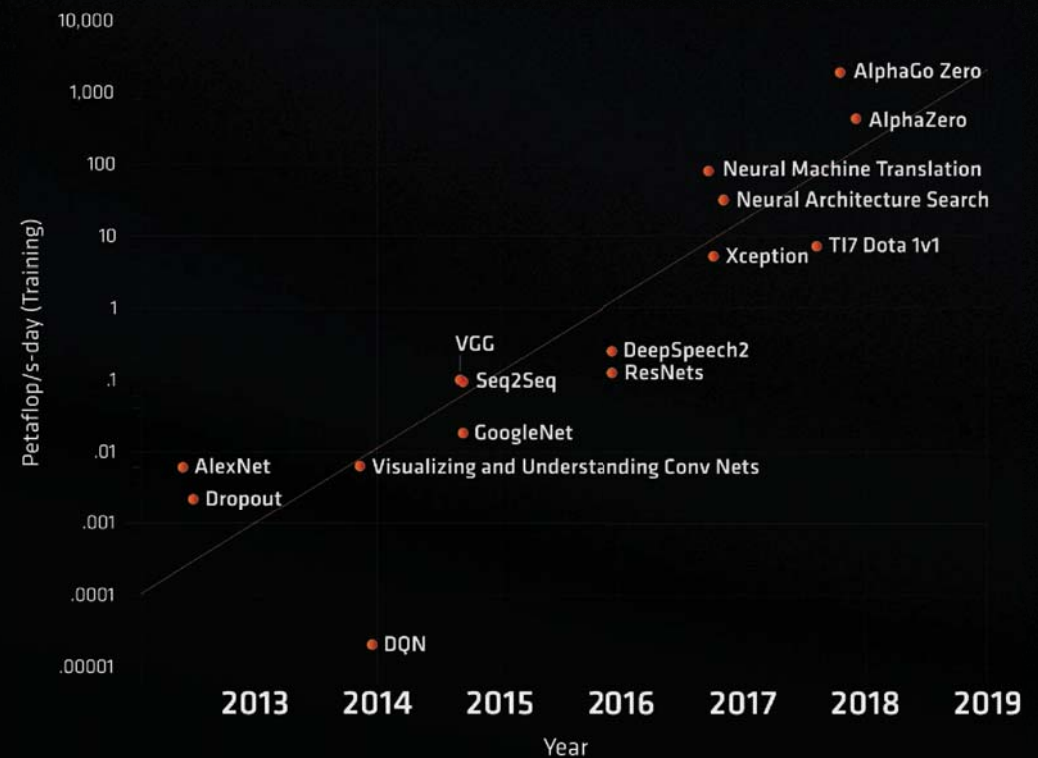
MACHINE LEARNING



REAL TIME
SIMULATION

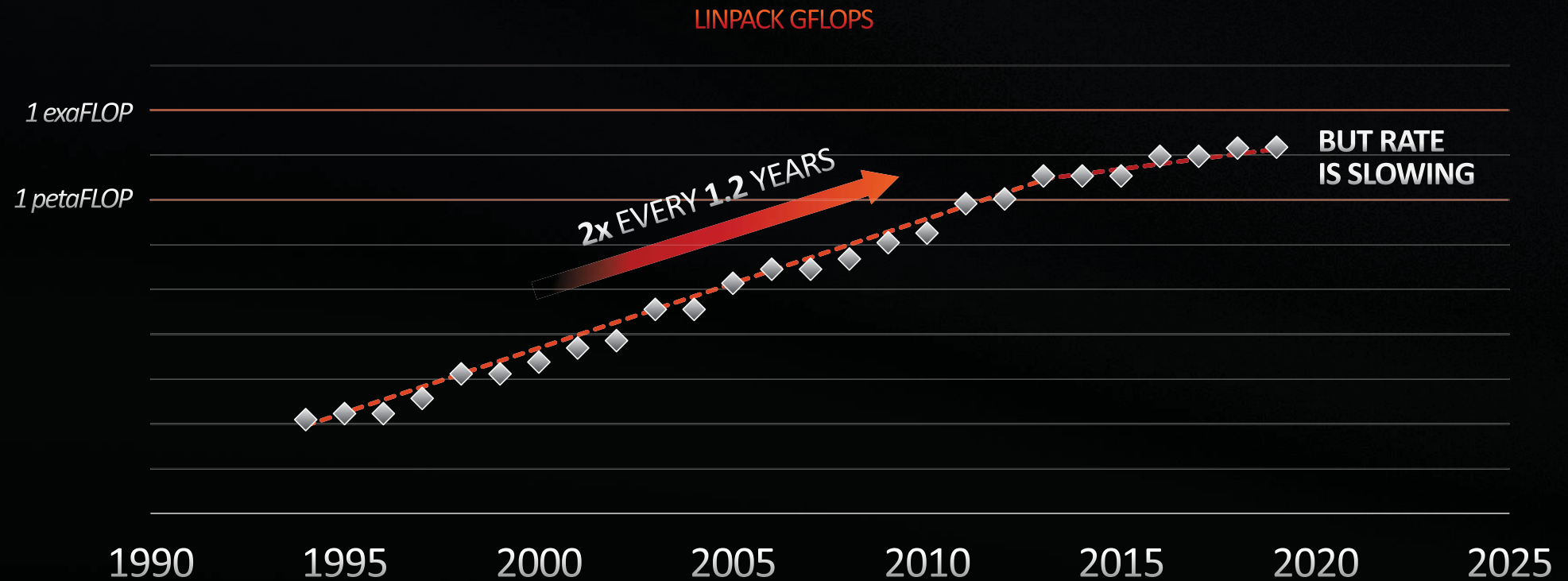
THE RELENTLESS DEMAND FOR MORE TRAINING

The advent of machine learning has driven a doubling of compute consumption every 3.5 months



THE RELENTLESS DEMAND FOR MORE COMPUTE

The World's Fastest Supercomputers



PRIMARY ELEMENTS OF HPC SYSTEM PERFORMANCE

Canonical Workload Runtime

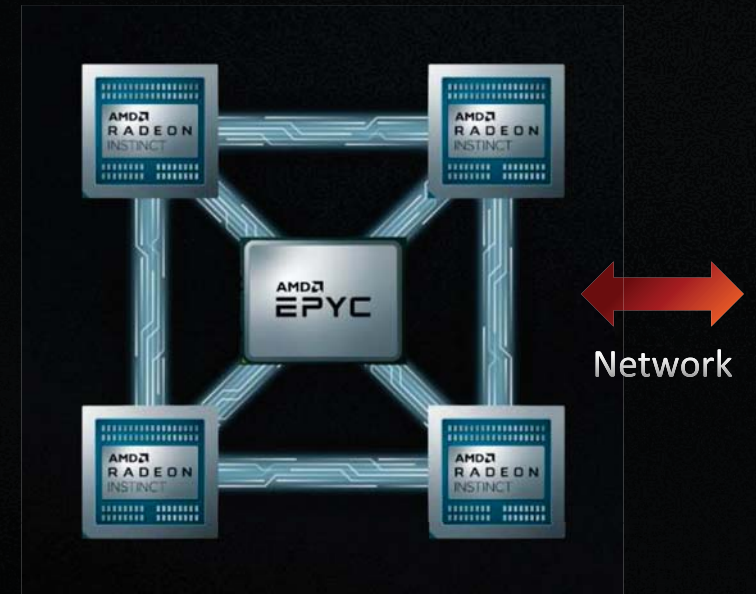
Network Bandwidth
and Latency

Intra-Node
Communication

CPU
Computation

GPU Computation

Memory Bandwidth
and Capacity



Common Workload Runtime Components

A horizontal stacked bar chart illustrating the distribution of time spent on different components. The x-axis represents the percentage of time, ranging from 0% to 100% in 10% increments. The bar is divided into four segments: GPU Computation (red, 53%), GPU Memory BW (orange, 12%), Intra-Node Connectivity (dark gray, 13%), and CPU Computation (light gray, 22%).

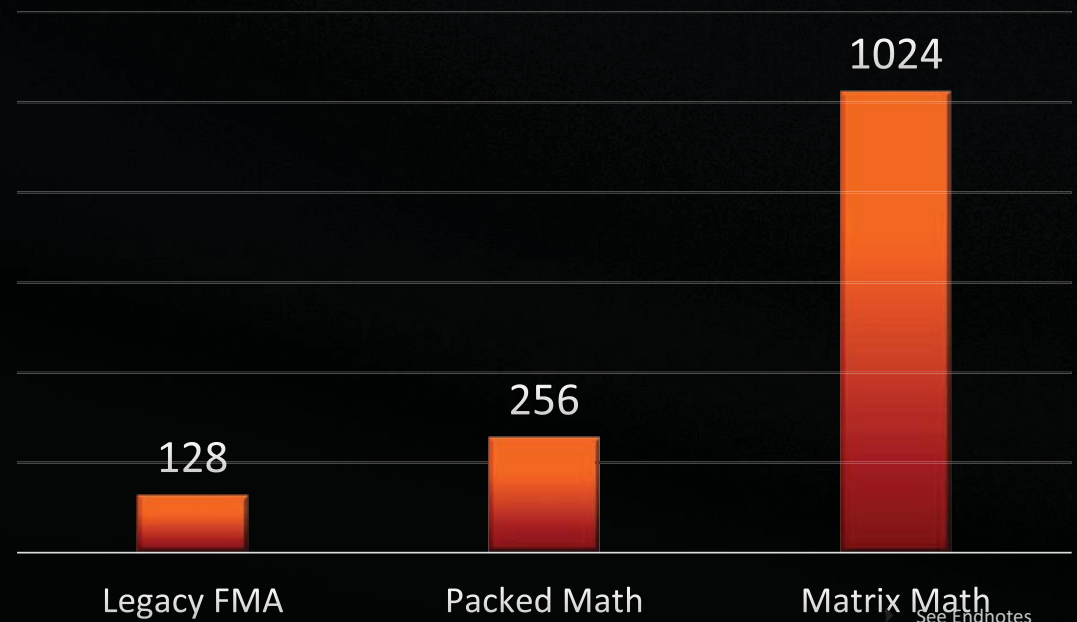
Component	Percentage
GPU Computation	53%
GPU Memory BW	12%
Intra-Node Connectivity	13%
CPU Computation	22%

Component	Percentage
CPU-GPU Connectivity	10%
CPU Computation	25%
GPU Memory BW	20%
GPU Computation	20%
GPU-GPU Connectivity	25%

OPTIMIZING EACH ELEMENT GPU COMPUTATION

- GPU compute FLOPS/Watt is a significant contributor to performance
- Improvement in GPU compute FLOPS/Watt is under pressure from slowdown in Moore's Law
- Architectural innovations are required to maintain historical trend

TRAINING FLOPS PER GPU COMPUTE UNIT

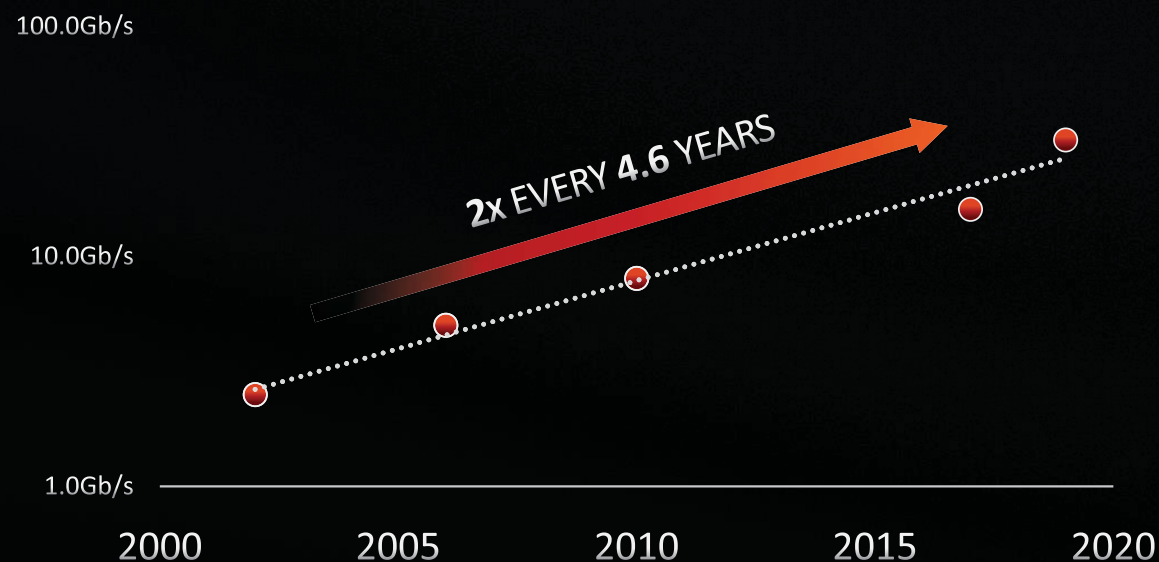


See Endnotes

OPTIMIZING EACH ELEMENT CONNECTIVITY CPU-GPU

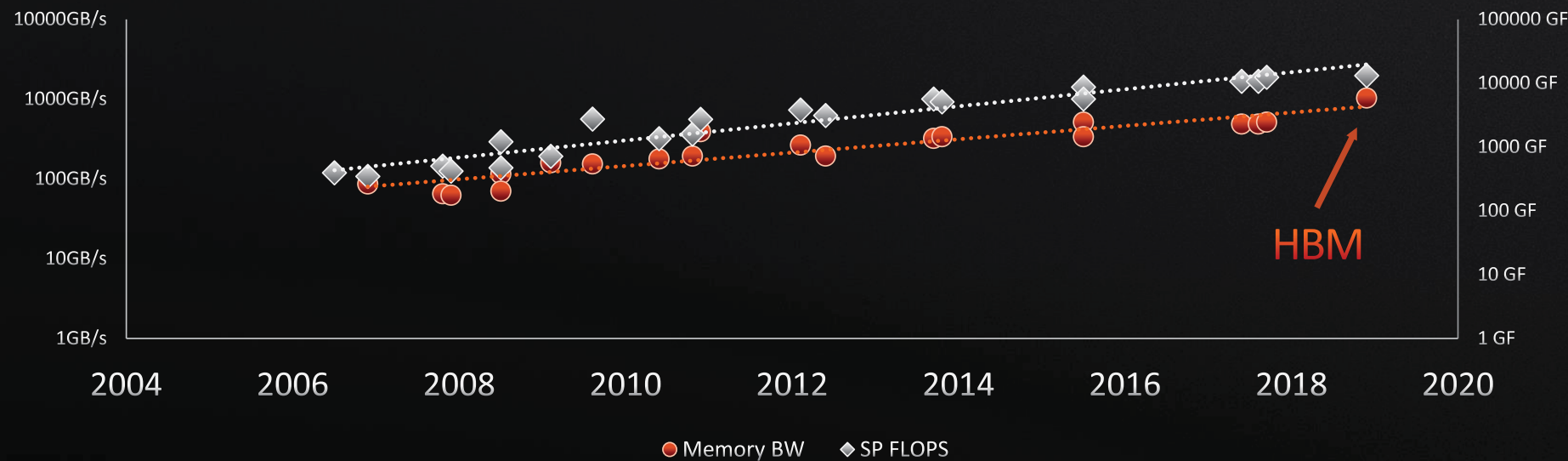
- High speed interfaces have been improving at a much slower rate than improvements in compute
- To support compute growth we must get more out of each bit of bandwidth

PCI EXPRESS SPEEDS



OPTIMIZING EACH ELEMENT MEMORY BANDWIDTH

GPU MEMORY BW OVER TIME



Memory BW is a Key Bottleneck

High Bandwidth 2.5D Memory
is Key Innovation

On-Die Cache Hierarchy
Must be Optimized



POWERING THE EXASCALE ERA

EPYC CPUs and Radeon Instinct GPUs to Deliver >1.5 exaFLOPS

Fully Optimized CPU and GPU
Design for Supercomputing

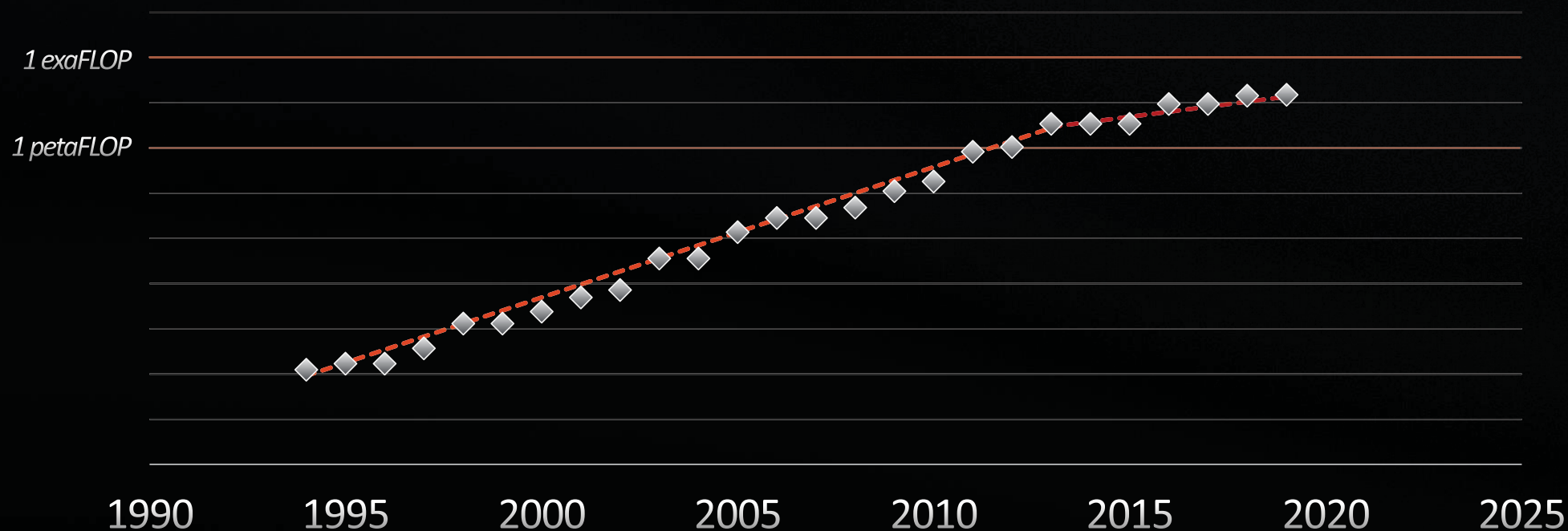
Leadership Interconnects
for System Performance

Open Software Tools to
Unlock the Performance

ACCELERATING HPC GAINS

WITH SYSTEM, SOFTWARE AND SILICON CO-OPTIMIZATION

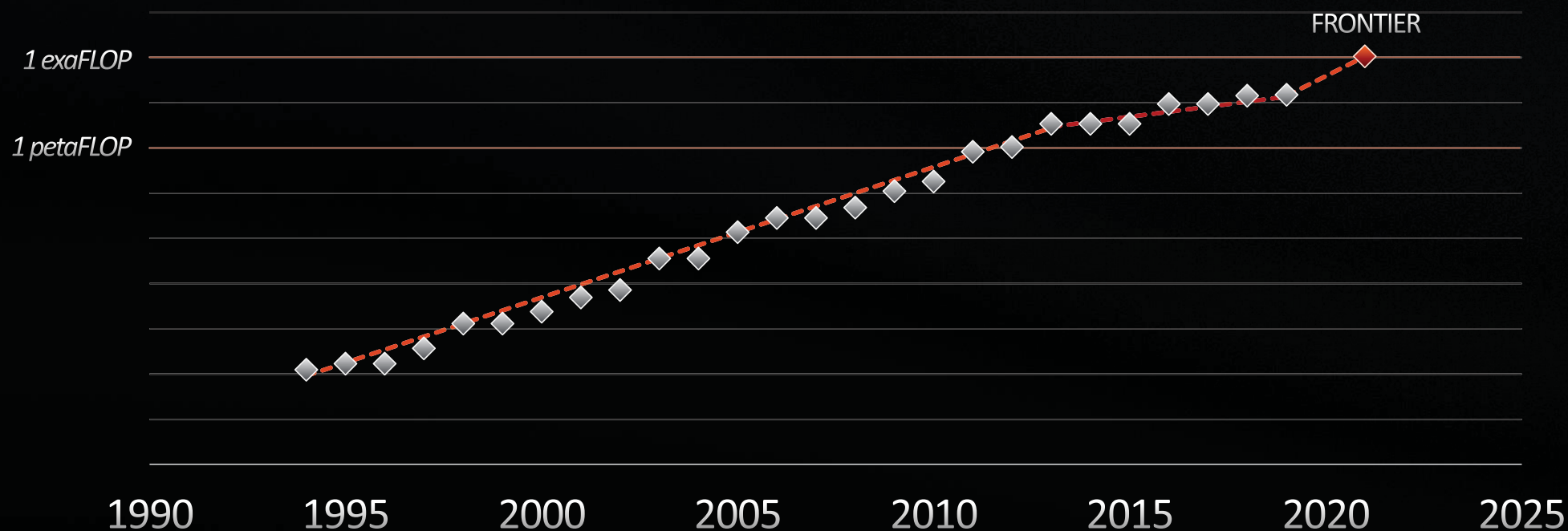
LINPACK GFLOPS



ACCELERATING HPC GAINS

WITH SYSTEM, SOFTWARE AND SILICON CO-OPTIMIZATION

LINPACK GFLOPS



DELIVERING HIGH PERFORMANCE COMPUTING FOR THE NEXT DECADE

Accelerated
Core IP

Chiplet
Architecture

High-Speed Coherent
Interconnects

System & Software
Co-Optimization

Continued
Technology Scaling

ACKNOWLEDGEMENTS

Dr. Lisa Su gratefully acknowledges the contributions of Vineet Goel, Frank Helms, Nick Malaya, Sam Naffziger, Allen Rush, Ben Sander, Swapnil Sakharshete, Mike Schulte in the development of this presentation.



ENDNOTES

Slides 4,5,6,9,10,11

Lisa T. Su, Samuel Naffziger, and Mark Papermaster, “Multi-Chip Technologies to Unleash Computing Performance Gains over the Next Decade,” IEDM Conference 2017.

Slide 11:

Original data up to the year 2010 collected and plotted by M. Horowitz, F Labonte O.Shacham, K. Olukotun, L. Hammond, and C. Batten.
New plot and data collected for 2010-2015 by K. Rupp. <https://www.karlrupp.net/2015/06/40-years-of-microprocessor-trend-data/>

Slide 17:

Testing by AMD Performance Labs as of 06/03/2019 utilizing 3rd Gen AMD Ryzen™ Processors: 3900X, 3800X, 3700X, 3600X, 3600 and Ryzen™ 7 2700X in Cinebench R20 1T.
Results may vary. RZ3-25
Based on June 8, 2018 AMD internal testing of same-architecture product ported from 14 to 7 nm technology with similar implementation flow/methodology, using performance from SGEMM. EPYC-07
Based on AMD internal testing, average per thread performance improvement at ISO-frequency on a 32-core, 64-thread, 2nd generation AMD EPYC™ platform as compared to 32-core 64-thread 1st generation AMD EPYC™ platform measured on a selected set of workloads including sub-components of SPEC CPU® 2017_int and representative server workloads. ROM-236

Slide 20

The comparison is based on the highest performing results for two-processor servers using AMD EPYC 7601 processors and Intel Xeon Gold 6248 processors published on www.spec.org as of April 27, 2019.

- • Score of 234 using 2 x Intel Xeon Gold 6248 processors.
- <https://www.spec.org/cpu2017/results/res2019q2/cpu2017-20190318-11225.html>
- • Score of 301 using 2 x AMD EPYC™ processor model 7601.
- <https://www.spec.org/cpu2017/results/res2019q1/cpu2017-20190304-11124.html>
- SPEC® and SPECrate® are registered trademarks of the Standard Performance Evaluation Corporation. Learn more at www.spec.org. NAP-170



ENDNOTES

Slides 20

A 2P EPYC™ 7742 processor powered server has SPECrate® 2017_int_base score of 682, <https://spec.org/cpu2017/results/res2019q3/cpu2017-20190722-16242.html> as of August 7, 2019. The next highest base score is a 2P Intel Platinum 9282 server with a score of 643, <http://spec.org/cpu2017/results/res2019q3/cpu2017-20190624-15369.pdf> as of July 28, 2019. SPEC®, SPECrate® and SPEC CPU® are registered trademarks of the Standard Performance Evaluation Corporation. See www.spec.org for more information. ROM-92

Slide 29

Dario Amodei and Danny Hernandez. "AI and Compute." <https://openai.com/blog/ai-and-compute/>.

Slide 31, 32, 35 AMD internal performance modeling and analysis

Slide 33 AMD Internal performance modeling and analysis

https://en.wikipedia.org/wiki/Graphics_Core_Next

Slide 38

<https://www.top500.org/> and ORNL performance estimate.